

Plain-Language Guide to the Hybrid EV Charging Experiment

Agentic AI-based Hybrid EV Charging Simulator - Joy Dutta

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Purpose of This Document

This document provides a plain-language explanation of the proof-of-concept EV-charging experiment reported in the IEEE Network magazine article *Agentic AI for Dynamic EV Charging: Explainable Coordination of Fixed Charging Stations and Mobile Charging Vehicles*. It summarizes what was simulated, why the experiment was designed this way, and how to interpret the reported results.

For step-by-step experiment reproduction (commands, file paths, and output artifacts), please see the technical companion document: `EXPERIMENT_RECREATION_TECHNICAL_GUIDE.PDF`.

The short idea is simple: a city has fixed fast-charging stations, but during peak hour some stations become busy. Mobile Charging Vehicles (MCVs) can be sent to station locations as temporary capacity boosters. Electric vehicles (EVs) arrive one by one, each driver has a different priority, and the operator must decide whether the EV should go to an FCS or use MCV support.

1 Real-World Story Behind the Simulation

Imagine a 20 km by 20 km city district during peak hour. People are returning from work, ride-hailing vehicles are active, some EVs are low on battery, and public fast chargers are not evenly loaded. Some charging stations have a queue, while others are farther away. The operator also owns a small fleet of battery backed mobile chargers. These mobile chargers can be sent from a central depot to the parking area of an existing charging station. In this experiment, each MCV is treated as a temporary capacity booster at an FCS location rather than as a random roadside charger.

This is similar to a real operational problem:

- A driver wants a good charging decision now.
- The operator wants to manage congestion and revenue.
- Fixed charging stations have limited ports.
- MCVs are flexible, but they have limited battery energy.
- Once an EV is assigned, that charger queue changes for the next EV.

The experiment asks whether an LLM-assisted coordination layer can make better or more policy-aware choices than a fixed deterministic rule, while still being bounded by the simulator's feasible actions.

2 What Physical System Was Simulated?

The simulation uses one shared physical setup:

- Region: 20 km by 20 km Manhattan-style service area.
- Road model: Manhattan distance, not straight-line Euclidean distance.
- Travel speed: 30 km/h.
- Depot: central MCV depot at coordinate (10,10) km.
- FCSs: 3 fixed charging stations.
- MCVs: 6 mobile charging vehicles.
- Base EV arrival pattern: one EV per minute for 10 minutes.
- Operator target: ideal wait below 5 minutes, acceptable wait below 10 minutes.

2.1 Fixed Charging Stations

Each FCS is modeled as a realistic modern DC fast-charging site:

Station	Location (km)	Ports	Power per port
FCS_1	(5, 5)	4	180 kW
FCS_2	(15, 5)	4	180 kW
FCS_3	(10, 15)	4	180 kW

At the start of the simulation, it is already peak hour. The stations are not empty. Each FCS has active sessions and some reserved/queued sessions. This is important because a newly arriving EV should not always see zero wait.

Station	Active remaining times (min)	Existing queued service times (min)
FCS_1	16, 20, 24, 28	22, 25, 28
FCS_2	8, 12, 16, 20	18
FCS_3	12, 18, 24, 30	24, 26

2.2 Mobile Charging Vehicles

The system has 6 MCVs. Each MCV is modeled as a battery-backed mobile charging unit:

Parameter	Value
Number of MCVs	6
Ports per MCV	2
Power per port	125 kW
Onboard energy	300 kWh
Minimum reserve	20 kWh
Depot recharge power	180 kW
Initial status	Idle at depot

An MCV cannot accept a new EV if serving that EV would reduce its onboard energy below 20 kWh. This prevents unrealistic behavior where a mobile charger keeps accepting EVs after its useful energy is effectively depleted.

3 What EVs Were Simulated?

The base experiment uses 10 EV requests. The vehicle mix was chosen to look like a realistic heterogeneous fleet rather than ten identical cars:

Brand	Model	Count	Usable battery	Avg. DC power
Tesla	Model Y Long Range AWD	2	75.0 kWh	124 kW
Tesla	Model 3 Long Range AWD	2	75.0 kWh	125 kW
Tesla	Model S Long Range	1	95.0 kWh	150 kW
BYD	Seal 82.5 kWh RWD Design	2	82.5 kWh	100 kW
BYD	ATTO 3	1	60.5 kWh	71 kW
Geely	EX5	1	60.2 kWh	95 kW
Volkswagen	ID.4 Pro	1	77.0 kWh	120 kW

Each EV also receives a random location inside the 20 km by 20 km region, a current state of charge between 14% and 38%, and a target state of charge between 72% and 84%. The required charging energy is calculated from the vehicle battery size and the gap between current and target state of charge.

4 What Do the EV Intents Mean?

The experiment uses three EV-side intents.

<code>time_optimized</code>	The driver mainly wants to minimize total time. Total time includes driving to the charger, waiting in the queue, and charging service time.
<code>cost_optimized</code>	The driver mainly wants to minimize money spent. This EV may accept more travel or more waiting if the charging cost is lower.
<code>greedy</code>	The driver chooses the currently best immediate tradeoff among time, price, distance, and MCV suitability. Greedy does not mean selfish in a moral sense. It means the EV does not plan for future system outcomes; it only chooses the best option visible now.

In the base 10-EV run, the intent mix is:

- 2 time-optimized EVs,
- 2 cost-optimized EVs,
- 6 greedy EVs.

5 What Does Operator Intent Mean?

The operator intent is the system-level policy. The experiment reported in the paper uses **revenue** mode. The codebase also contains other policy modes for future experiments.

<code>revenue</code>	The operator protects revenue and treats MCV service as premium mobile capacity. It avoids unnecessary MCV deployment unless congestion is severe.
<code>peak_resilience</code>	The operator is more aggressive about clearing queues and reducing waiting time. MCVs are encouraged as congestion relief tools.

equity The older single-EV project includes this mode as a fairness-oriented policy. In the multi-fleet proof-of-concept, the reported experiments use **revenue**; equity-like policies can be added through policy knobs, feasibility filters, or satisfaction weights.

The current multi-fleet simulation does not rely on one static **OPERATOR_SITUATION_DESCRIPTION**. Instead, it generates operator context dynamically for every EV arrival using current FCS waits, MCV status, the active policy mode, policy knobs, and the EV’s situation.

6 Why Three Main Controllers?

The extended evaluation compares three controllers because each one answers a different research question.

Controller	What it does	Why it is included
deterministic_fixed	Uses the original deterministic ranking rule. No LLM calls.	This is the simple baseline. It shows how the original fixed rule behaves.
deterministic_satisfaction	Provides the same feasible option list as the EV LLM and directly chooses the option with maximum estimated satisfaction. No LLM calls.	This is a strong non-LLM comparator. It prevents a weak comparison where the LLM only beats a simple rule.
llm_full	Uses one bounded Operator LLM call and one bounded EV LLM call per EV arrival.	This tests the proposed agentic architecture: the LLM interprets intent and chooses only among feasible simulator-generated actions.

The important point is that the LLM is not allowed to invent resources. It only sees feasible options that the simulator has already calculated.

7 Normal Evaluation

The normal comparison is the simple 10-EV experiment.

How to Reproduce the Run

Step-by-step reproduction instructions (commands, file paths, and output artifacts) are provided in **EXPERIMENT_RECREATION_TECHNICAL_GUIDE.tex**.

Normal Evaluation Result

The reported normal comparison produced:

Metric	Deterministic baseline	LLM-assisted run
Average satisfaction	85.8/100	96.8/100
Average wait	12.47 min	2.47 min
EVs compared	10	10
Accepted EV LLM picks	N/A	10/10
Choices by resource	10 FCS	10 FCS

In plain words, the LLM-assisted run improved average EV satisfaction by 11.0 points and reduced average waiting time by about 10 minutes per EV in this normal comparison. It did this while still choosing only feasible options.

8 Extended Evaluation

The extended evaluation was created for the paper revision because reviewers asked for clearer empirical support, stronger baselines, boundedness evidence, and a more credible hybrid FCS–MCV stress test.

How to Reproduce the Run

Detailed reproduction instructions for the extended evaluation (how to run it, where outputs are written, and how to verify the boundedness logs) are provided in `EXPERIMENT_RECREATION_TECHNICAL_GUIDE.tex`.

What the Extended Evaluation Does

The extended evaluation has three stages.

1. **Demand sweep.** It runs EV counts from 10 to 30 in steps of 2, with one EV arriving every 60 seconds.
2. **Fallback stress stage.** If no MCV activity appears by 30 EVs, the evaluator can compress arrivals into a burst or apply a mild FCS outage. In the completed experiment, this fallback was not needed.
3. **Representative reruns.** It reruns routine, hybrid-threshold, and strong-stress scenarios with three seeds for cleaner reporting.

Threshold Result

The extended evaluation found:

Quantity	Result
First EV count where an MCV was selected	10
First EV count where an MCV was dispatched	10
Hybrid-threshold scenario	<code>stage1_ev10_ia60</code>
Strong-stress scenario	<code>stage1_ev30_ia60</code>
Fallback burst needed?	No
Fallback outage needed?	No

This means that hybrid behavior was visible even in the 10-EV setting. The routine and hybrid-threshold cases therefore collapse to the same physical load level in this run.

9 How to Read the Extended Results

Column	Meaning
avg_satisfaction	Average EV user satisfaction on a 0–100 scale.
avg_wait_min	Average queueing delay after the EV reaches the charger.
avg_total_time_min	Travel time plus waiting time plus charging time.
avg_cost_usd	Average charging cost paid by the EV.
fcs_selected_count	Number of EVs assigned to FCS options.
mcv_selected_count	Number of EVs assigned to MCV options.
mcv_dispatch_count	Number of MCV deployment actions made by the operator.
total_llm_calls	Total Operator plus EV LLM calls.
feasible_choice_rate	Fraction of EV LLM choices that stayed inside the feasible set.

10 Representative Extended Results

Scenario	Controller	Sat.	Wait	Total	MCV sel.	Calls
10 EVs	Fixed	85.0	12.52	44.42	0.00	0
10 EVs	Satisfaction-aware	94.9	0.69	43.49	2.67	0
10 EVs	LLM full	96.7	1.57	46.03	0.00	20
30 EVs	Fixed	80.5	15.08	47.80	3.67	0
30 EVs	Satisfaction-aware	94.3	0.50	43.48	13.00	0
30 EVs	LLM full	97.5	5.61	54.07	2.33	60

These numbers should be read carefully. The satisfaction-aware deterministic controller is very strong because it directly optimizes the satisfaction score. It often reduces waiting time by using MCVs aggressively. The LLM-assisted controller does not always minimize wait, but it achieves high satisfaction while using fewer MCV assignments under the revenue policy. That is the main selling point: the LLM layer is not just a wait minimizer. It is a bounded policy-aware coordinator.

11 What the Results Signify

The original fixed deterministic baseline is useful but narrow. It follows a fixed score and can miss a better satisfaction tradeoff.

The deterministic satisfaction-aware controller is the strongest non-LLM baseline. It shows what happens when the system directly maximizes the same satisfaction metric used for evaluation. This is important because it makes the LLM comparison fairer. If the LLM still performs well against this controller, then the LLM is not only beating a weak heuristic.

The LLM-assisted controller shows the proposed architecture. The simulator does the physics and feasibility. The LLM reads the operator and EV context, then chooses only from valid options. This makes the LLM useful as an explanation and policy-translation layer, not as an unconstrained controller.

12 Boundedness and Safety

The extended evaluation audited every LLM EV decision. The completed run produced:

- 490 audited LLM EV decisions,
- 490 choices inside the feasible set,

- 0 invalid LLM EV selections,
- 0 fallback events.

Across the full extended package, the total LLM call count was:

Part	Total calls	Operator calls	EV calls
Demand sweep	440	220	220
Representative reruns	540	270	270
Total	980	490	490

This is why the experiment is called bounded LLM-assisted coordination. The LLM is called sparsely at decision events. It is not controlling power electronics or real-time charger hardware.

13 Main Takeaway

The experiment supports the proof-of-concept claim in the paper:

An LLM can be used as a bounded, explainable coordination layer above a deterministic EV charging simulator. It can interpret EV and operator intent, select among feasible FCS/MCV options, and produce auditable decisions without inventing physical resources or bypassing feasibility checks.

The deterministic satisfaction-aware controller is an excellent benchmark, but it is still a fixed mathematical rule. The LLM-based architecture is useful when the operator needs policy interpretation, human-readable reasoning, stakeholder alignment, and an audit trail in addition to numerical optimization.